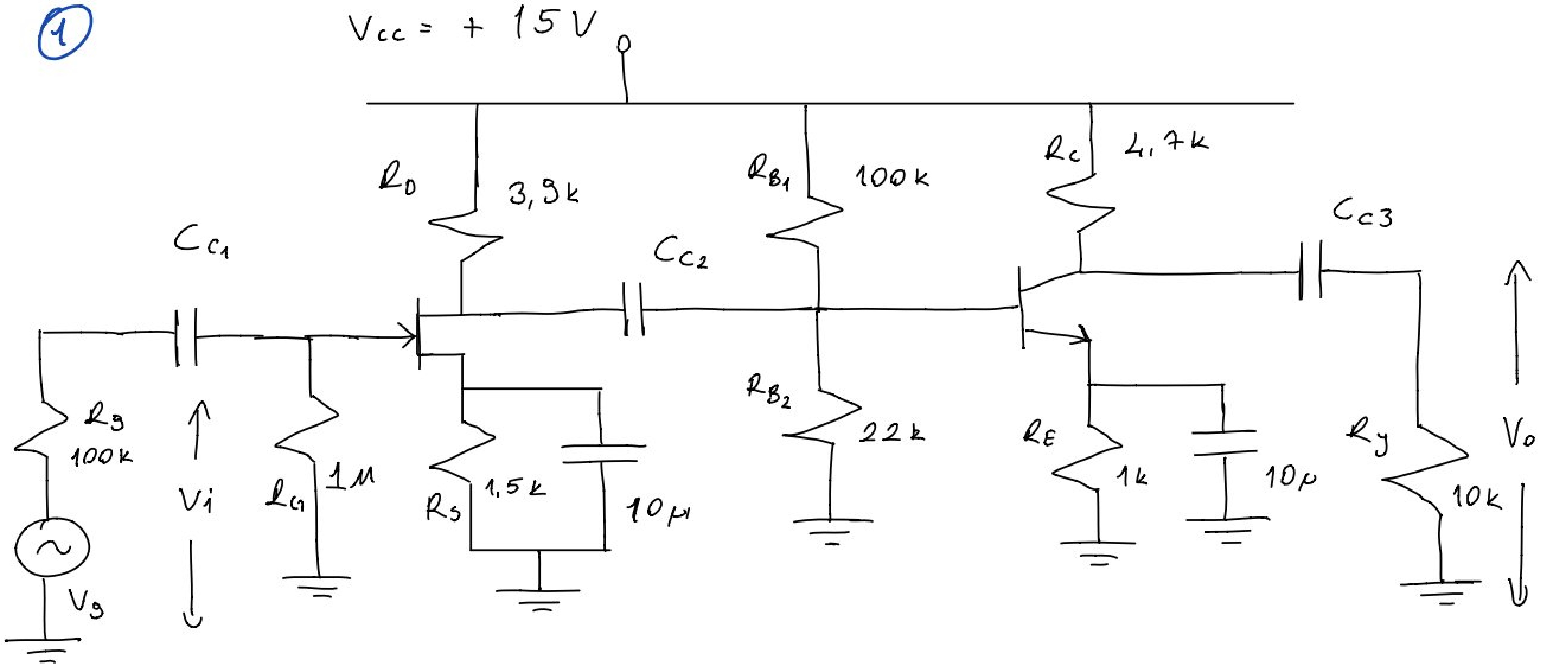
1



$I_{DSS} = 6mA$, $V_p = -3V$, $\beta = h_{fe} = h_{FE} = 200$, $V_{BE} = 0,7V$

$1/h_{oe} = 50k$ $\quad C_{gs} = 3pF$, $C_{ds} = 1pF$, $C_{gd} = 3pF$

$C_{be} = 80pF$, $C_{bc} = 3pF$, $C_{ce} = 6pF$

a) $K_v = \dfrac{V_o}{V_i} = ?$

b) Alt kesim frekansının 10 Hz olabilmesi için, C_{c1}, C_{c2}, C_{c3} = ?

c) Üst kesim frekansı nedir?

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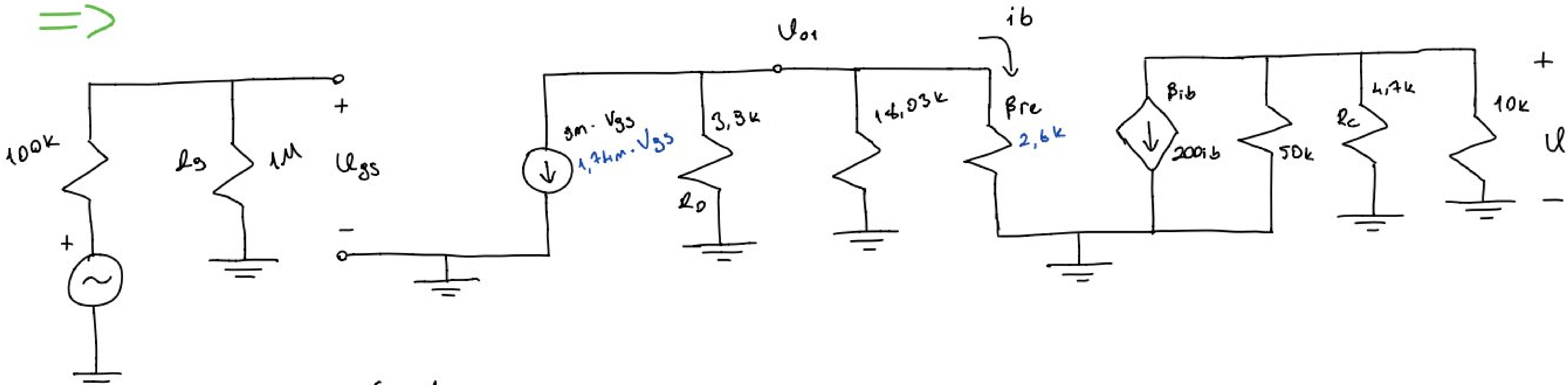


a)

$$g_m = 2\,\frac{6mA}{3}\,(1-0,565) = 1,74m$$

$$r_e = \frac{26mV}{I_{CQ}} \qquad V_B = 15\cdot\frac{22k}{122k} = 2,7V \qquad V_E = 2,7-0,7 = \underline{2V}$$

$$I_{CQ} \cong I_{EQ} \qquad I_{EQ} = \frac{2}{1k} = 2mA$$

$r_e = 13\Omega$

$h_{ie} = 2,6k\Omega$

$$A_{v1} = -g_m(3{,}9k // R_{i2}) \qquad R_{i2} = 18{,}03k // 2{,}6k = 2{,}272k$$

$$A_{v1} = -2{,}498$$

$$A_{v2} = \frac{-200}{2{,}6k}\left(\underbrace{50k // 4{,}7k // 10k}_{3{,}005k}\right) = -231{,}15$$

$$A_v = A_{v1} \cdot A_{v2} = 577{,}41$$

b)

$$C_{c1} \Rightarrow \quad 10 = \frac{1}{2 \cdot \pi \cdot C_{c1} \cdot (100k + 1M)} \Rightarrow C_{c1} = 14{,}46\ nF$$

$$C_{c2} \Rightarrow \quad 10 = \frac{1}{2\pi \cdot C_{c2} \cdot (2{,}27k + 3{,}9k)} \Rightarrow C_{c2} = 2{,}57\ \mu F$$

$$C_{c3} \Rightarrow \quad 10 = \frac{1}{2 \cdot \pi \cdot C_{c3}\ (4{,}7k + 10k)} \Rightarrow C_{c3} = 1{,}48\ \mu F$$

c)

— FET —

$$C_{in} = C_{gs} + C_{gd}(1 - A_{v1}) + C_{w1}$$

$$C_{in} = 3pF + 3pF(1 + 2{,}498) + 0$$

$$C_{in} = 13{,}49pF \qquad R_{in} = 1M // 100k \cong 90{,}9\ k\Omega$$

$$f_{cin} = \frac{1}{2 \cdot \pi \cdot 13{,}43 \times 10^{-12} \cdot 90{,}9 \cdot 10^{3}} = 129{,}730\ kHz$$

$$C_{out} = C_{ds} + C_{gs}\left(1 - \frac{1}{A_{v1}}\right) + C_{w2}$$

$$C_{out} = 1pF + 3pF\left(1 + \frac{1}{2{,}498}\right) + 0 \cong 5{,}2\ pF$$

$$R_{out} = 3{,}9k // 2{,}27 = 1{,}434\ k\Omega$$

$$f_{cout} = \frac{1}{2 \cdot \pi \cdot 5{,}2 \times 10^{-12} \cdot 1434} = 21{,}343\ MHz$$

— BJT —

$$C_{in} = C_{be} + C_{bc}(1 - A_{v2}) + C_{w1}$$

$$C_{in} = 80pF + 3pF(1 + 231{,}15) + 0 \cong 776{,}4pF$$

$$R_{out} = 3{,}9k // 2{,}27 = 1{,}434\ k\Omega$$

$$f_{cin} = \frac{1}{2 \cdot \pi \cdot 776{,}4 \times 10^{-12} \cdot 1434} = 143\ kHz$$

$$C_{out} = C_{ce} + C_{bc}\left(1 - \frac{1}{A_{v2}}\right) + C_{w1}$$

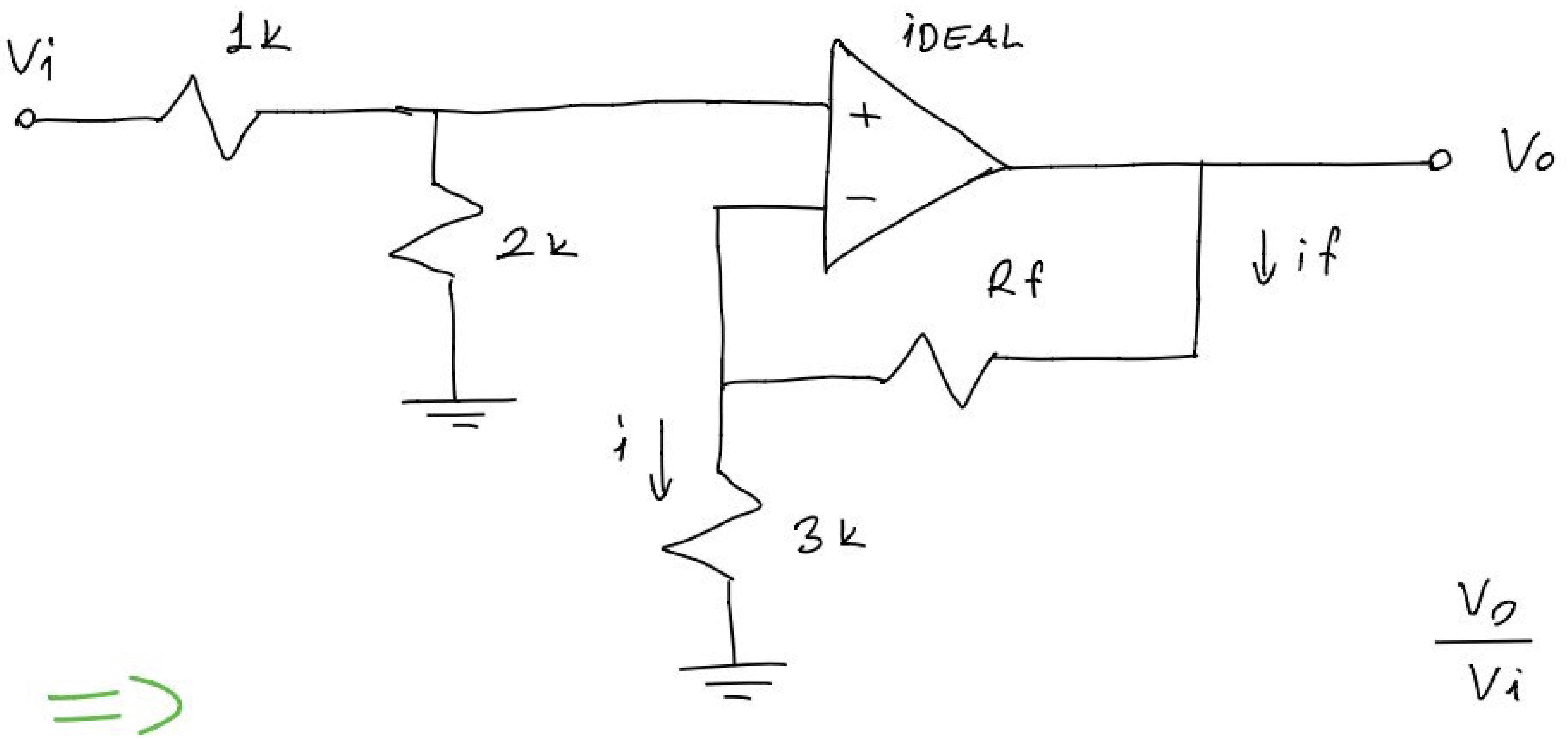
$$C_{out} = 6pF + 3pF\left(1 + \frac{1}{231{,}15}\right) + 0 \cong 9{,}02pF$$

$$R_{out} = 50k // 4{,}7k // 10k = 3{,}005\ k\Omega$$

$$f_{cout} = \frac{1}{2 \cdot \pi \cdot 9{,}02 \cdot 10^{-12} \cdot 3005} = 5{,}87\ MHz$$

$$f_{üst} = 129{,}730\ kHz$$

② Aşağıda verilen ideal OPAMP'ın kazancının 5 olabilmesi için R_f ne olmalıdır?



$$\frac{V_o}{V_i} = 5$$

$\Rightarrow$

Gerilim Bölücü

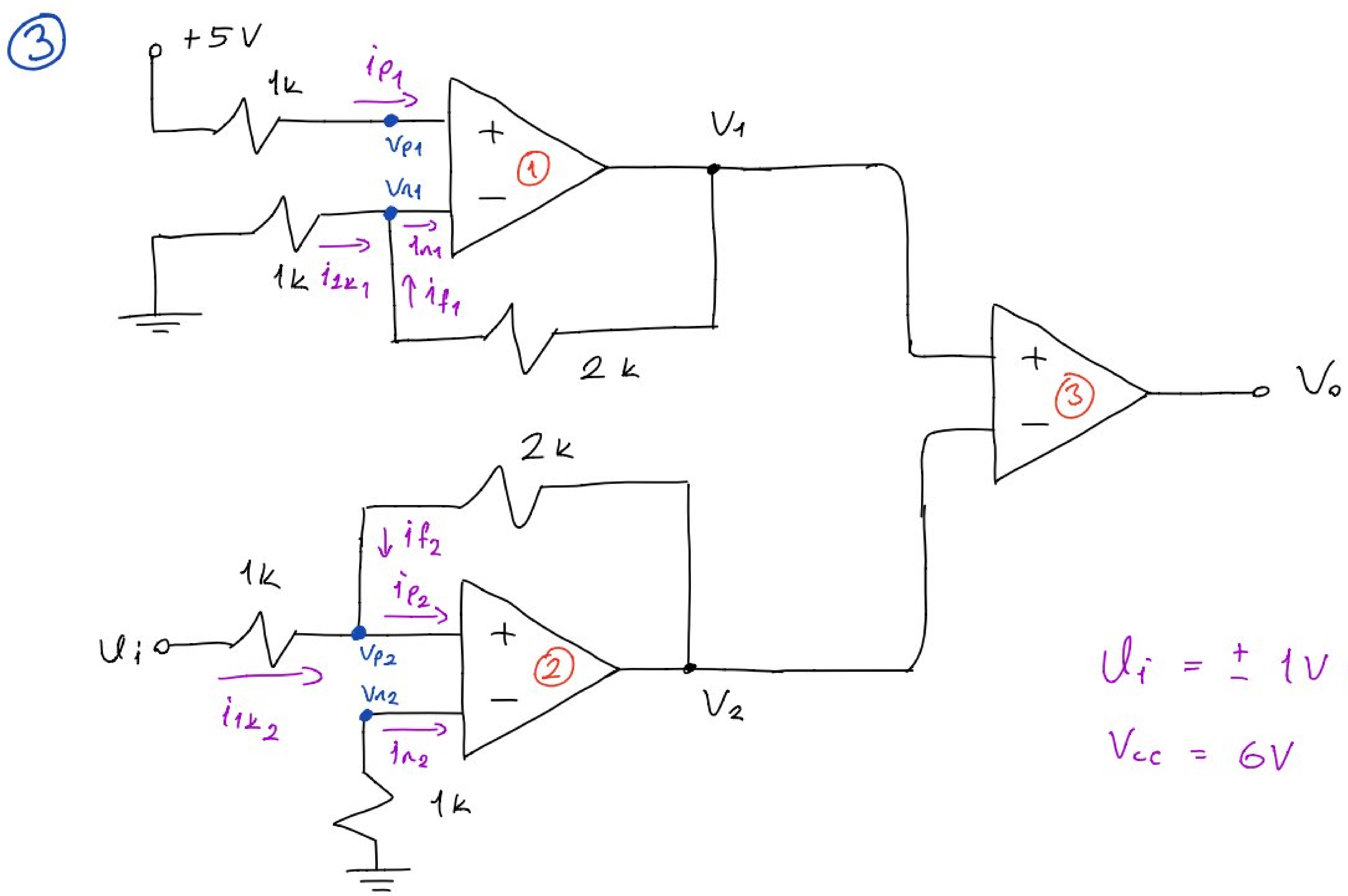
$$V_s = V_i \frac{2k}{1k + 2k} = \frac{2V_i}{3}$$

$$\frac{V_o}{V_s} = \frac{V_o}{\frac{V_i \cdot 2k}{3k}} \Rightarrow \frac{3}{2} \cdot \frac{V_o}{V_i} = \frac{3.5}{2} = \frac{15}{2}$$

$$\frac{V_o}{V_s} = 1 + \frac{R_f}{R_i} \Rightarrow \frac{15}{2} = 1 + \frac{R_f}{3k} = \frac{39k}{2}$$

$$\Rightarrow \quad R_f = 19{,}5\,k\Omega$$

③



$U_i = \pm 1V$ (sinüzoidal)

$V_{cc} = 6V$

Şekildeki devrede U_i, $\pm 1V$ sinüzoidal bir singal ve tüm Opamlar için $\pm V_{cc} = \pm 6V$ olduğuna göre, V_1, V_2 ve V_o singallerini U_i'ye göre ölçekli olarak çiziniz.

=>

1. OP-AMP

$V_{p_1} = V_{n_1} = 5V$

$i_{p_1} = i_{n_1} = 0A$

$i_{1k_1} + i_{f_1} = 0A$

$$\frac{V_{n_1}}{1k} + \frac{V_{n_1} - V_1}{2k} = 0$$

$10 + 5 - V_1 = 0 \Rightarrow V_1 = 15V$ ($V_{cc} = \pm 6V$ ise)

$\Rightarrow \underline{V_1 = 6V}$

2. OP-AMP

$i_{p_2} = i_{n_2} = 0A$

$V_{p_2} = V_{n_2} = 0V$

$i_{1k_2} + i_{f_2} = 0A$

$$\frac{V_i - \overbrace{V_{p_2}}^{0V}}{1k} + \frac{V_2 - V_{p_2}}{2k} = 0 \Rightarrow \frac{V_i}{1k} = \frac{-V_2}{2k}$$

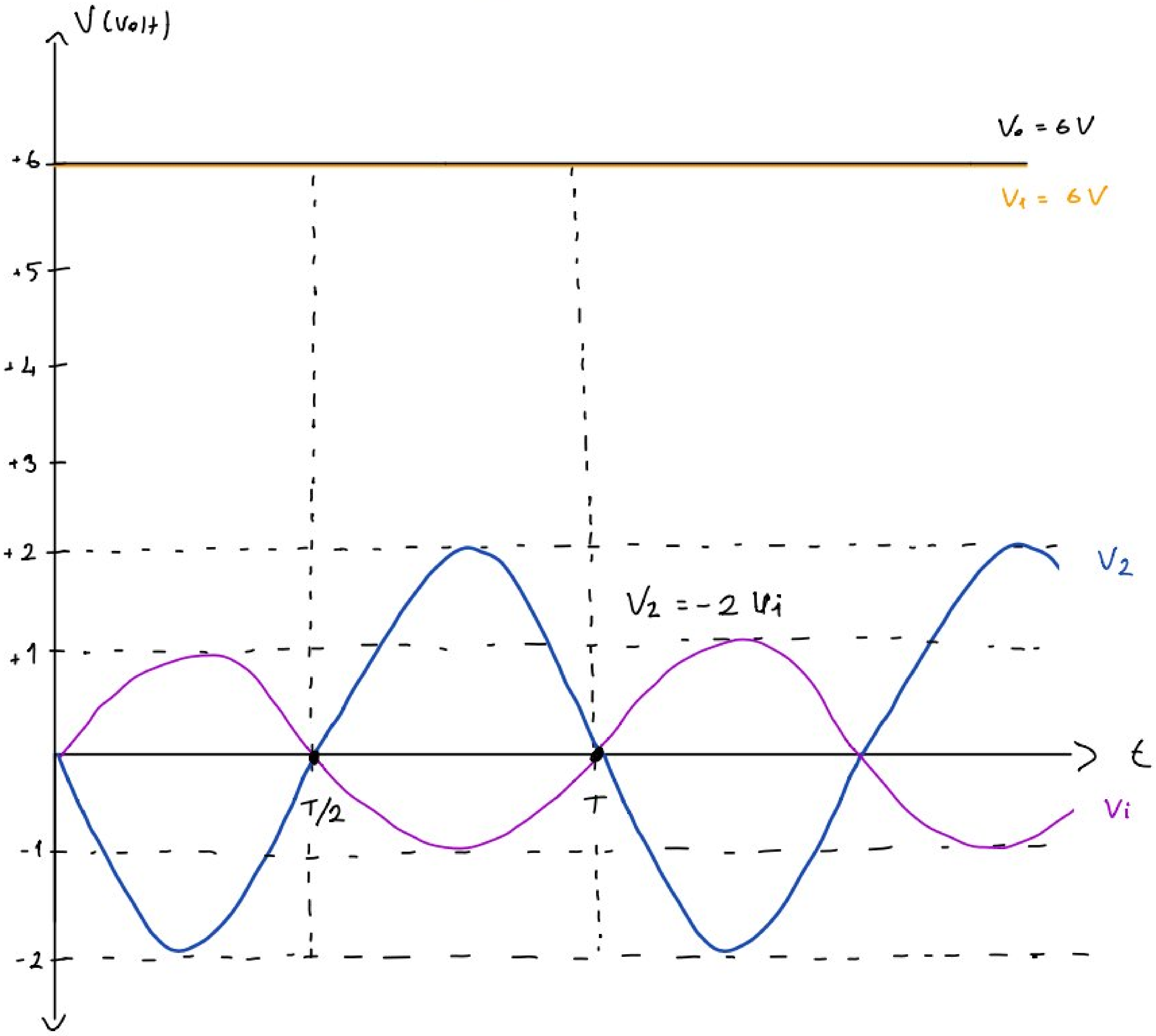
$\underline{V_2 = -2V_i \ V}$ ($V_i = \pm 1V$ (sinüzoidal))

$\Rightarrow \underline{V_2 = -2V \text{ (sinüzoidal)}}$

3. OP-AMP

$V_1 > V_2$ olduğundan dolayı

$$\underline{\underline{V_0 = 6V}}$$



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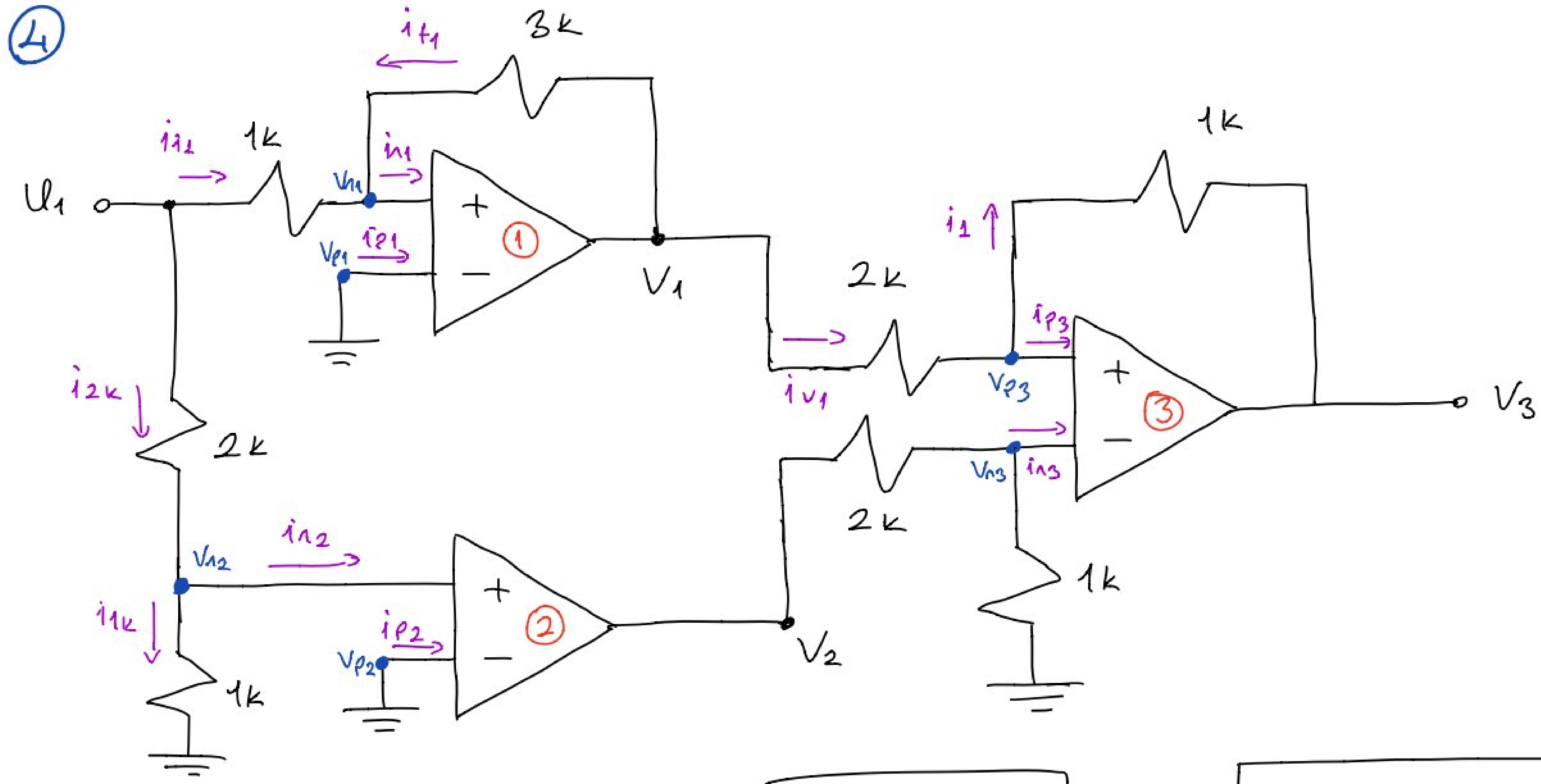
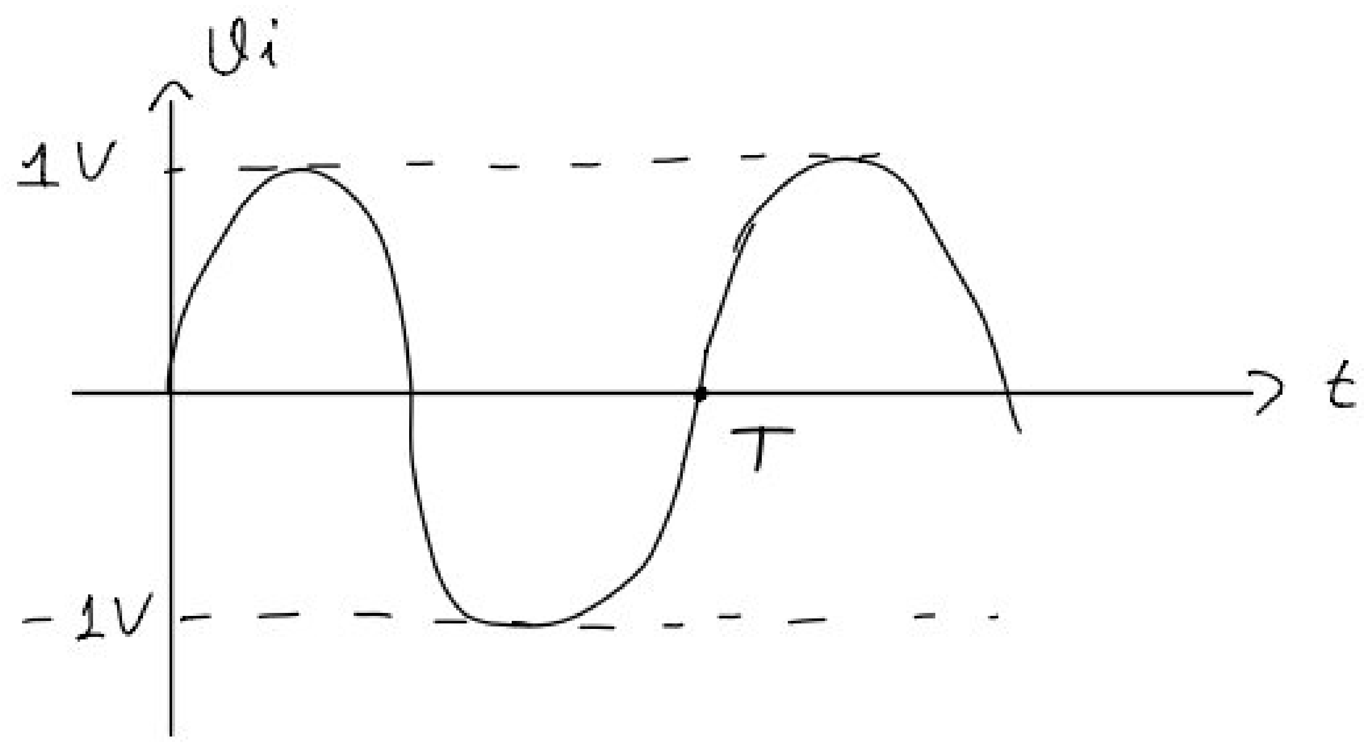


$\boxed{+\ V_{cc} = \pm 12V}$ $\boxed{V_i = \pm 1\,V \text{ (sin)}}$



Buna göre $V_1,\ V_2,\ V_3 = ?$

(çiziniz)

$\Rightarrow$

1. OP-AMP

$$\left.\begin{aligned} V_{p1} &= V_{n1} = 0V \\ i_{n1} &= i_{p1} = 0A \\ i_{i1} + i_{f1} &= 0 \end{aligned}\right\}$$

$$\frac{\overbrace{V_{n1}}^{0V} - V_i}{1k} + \frac{\overbrace{V_{n1}}^{0V} - V_1}{3k} = 0 \Rightarrow \underline{\underline{V_1 = -3V_i}}$$

2. OP-AMP

$$\left.\begin{aligned} V_{n2} &= \frac{V_i \cdot 1k}{2k + 1k} = \frac{V_i}{3} \\ V_{p2} &= 0V \end{aligned}\right\} V_{n2} > V_{p2} \Rightarrow \quad V_2 = \begin{cases} 12V, & V_{n2} < V_{p2} \\ 0V, & V_{n2} = 0V \\ -12V, & V_{n2} > V_{p2} \end{cases}$$

3. OP-AMP

$$V_{n3} = \frac{V_2 \cdot 1k}{2k + 1k} = \frac{V_2}{3}$$

$$i_1 = i_{v1}$$

$$V_{n3} = V_{p3}$$

$$\frac{\overbrace{V_{p3}}^{V_2/3} - V_3}{1k} = \frac{V_1 - \overbrace{V_{p3}}^{V_2/3}}{2k}$$

$$\Rightarrow \frac{V_2}{3} - V_3 = \frac{V_1}{2} - \frac{V_2}{6}$$

$$\Rightarrow V_3 = \frac{V_2 - V_1}{2}$$